

- 1) Define cloud computing and explain any four characteristic that makes it different from traditional on-premise computing.

Definition

Cloud computing is the delivery of computing service such as servers, storage, databases, networking and software over the internet on a pay as-you-use basis.

Characteristics

1) On Demand Self-Service:

User can access computing resources whenever needed without human intervention.

2) Resource Pooling:

Resources are shared among multiple users using virtualization.

3) Rapid Elasticity

Resources can be increased / decreased quickly based on demand.

4) Measured Service:

User pay only for the resource they consume (pay-as-you-go).

- 2) Trace the evolution of cloud computing from hardware evolution to internet software evolution and show how these changes enabled modern cloud platform.

Hardware Evolution

Powerful servers, multi-core processor and large storage system improved computing performance.

Virtualization

Hypervisors enabled multiple virtual machines to run on one physical server, improving resource utilization.

Distributed Computing:

Multiple computers worked together to process load workload.

Internet Evolution:

High-speed internet enabled remote access to computing resources.

Web Service

APIs, SOAP and REST allowed application to communicate over the internet.

Cloud Platform:

These techniques led to modern cloud services such as AWS, Microsoft Azure and Google, offering scalable and on-demand computing.

3) Explain the role of service virtualization in cloud computing. Difference between virtualization as an enabling technology and cloud computing as a service model.

Role of service virtualization:

- * Create multiple virtual machines on one physical server.
- * Improves hardware utilization.
- * Reduces infrastructure cost.
- * Provides isolation between applications
- * Enables scalability and disaster recovery.

Difference:

Virtualization	Cloud Computing
Technology that creates virtual machine.	Service model that delivers IT resources over the internet
Uses hypervisor	Uses virtualization plus networking and automation
Focuses on efficient hardware usage	Focuses on delivering service (IaaS, PaaS, SaaS)
Runs within a server	Accessible from anywhere via the internet.

4. Discuss the role of data centers in delivery cloud services.

Data centers are facilities that contain server storage devices, networking equipment power supply and cooling system.

Role :-

- * Host cloud application and service
- * Store and process cloud data.
- * Provide high availability and reliability
- * Ensures data security and backup.
- * Support scalability by adding resources when demand increases.
- * Enable disaster recovery and business continuity.

5) Differentiate IaaS, PaaS, SaaS and XaaS with one suitable real-world example for each model.

Service Model	Description
IaaS (Infrastructure as a service)	Provide virtual service, storage and network.
PaaS (Platform as a service)	Provide a platform for developing and deploying application
SaaS (software as a service)	Provides ready-to-use software over the internet
XaaS (Anything as a service)	Delivery any IT service through the cloud.